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Image acquisition apparatus and self-moving device

Abstract

An image acquisition apparatus and a self-moving device. The image acquisition apparatus includes a base; an upper box body, the upper box body is detachably disposed on the base, a cavity is formed between the upper box body and the base, and a front end face of the upper box body is provided with a window; a camera assembly, the camera assembly is fixedly disposed on the base and is placed in the cavity, and the camera assembly and the window are correspondingly provided; a sunshading part, the sunshading part is disposed at a side of the upper box body away from the base, and the sunshading part is disposed spaced from the upper box body.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority to Chinese Patent Application No. 202222116678.9, filed on Aug. 11, 2022, which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

(2) The present application relates to the field of image acquisition technology and, in particular, to an image acquisition apparatus and a self-moving device.

BACKGROUND

(3) When some large machines such as harvesters and mowers work outdoors, the situation of visual blind spots may occur, therefore an image acquisition apparatus needs to be installed to assist the machines to work. Because the image acquisition apparatus, when working, needs to perform data processing on the pictures acquired by the image acquisition apparatus itself, the data processing volume is large, the heat generation is serious. In addition, the image acquisition apparatus cannot avoid direct sunlight when working outdoors, which also causes temperature rise of the image acquisition apparatus and even device damage.

(4) The problem of high temperature rise of the existing image acquisition apparatus cannot be solved by the structure of the image acquisition apparatus, which leads to reduced working efficiency of the image acquisition apparatus.

SUMMARY

(5) In order to solve the aforesaid problem, the present application provides an image acquisition apparatus to solve the problem of high temperature rise of the image acquisition apparatus. According to the first aspect of the present application, it is provided an image acquisition apparatus including: a base; an upper box body, the upper box body is detachably disposed on the base, a cavity is formed between the upper box body and the base, and a front end face of the upper box body is provided with a window; a camera assembly, the camera assembly is fixedly disposed on the base and is placed in the cavity, and the camera assembly and the window are correspondingly provided; a sunshading part, the sunshading part is disposed at a side of the upper box body away from the base, and the sunshading part is disposed spaced from the upper box body.

(6) Optionally, a surface of the sunshading part near the upper box body is provided with a plurality of bolts, and a surface of the upper box body near the sunshading part is provided with a plurality of bolt holes.

(7) Optionally, a lower surface of the base is provided with a wiring hole for a wire harness to pass through, and a rubber plug is disposed between the wire harness and the wiring hole.

(8) Optionally, an upper surface of the base is provided with a circle of closed groove, and an inner surface of the upper box body is vertically provided with a circle of closed first rib, and the first rib is embedded in the groove for interference fit. The first rib, the groove, the base, and the upper box body enclose to form a first cavity, and the camera assembly is placed in the first cavity.

(9) Optionally, a waterproof ring is disposed in the groove, and an upper surface of the first rib is

pressed against the waterproof ring.

(10) Optionally, the upper box body further includes a transparent part, and the transparent part is embedded on the window.

(11) Optionally, the camera assembly includes a control board, a camera bracket, and a camera; the control board is fixedly disposed on the base, the camera bracket is vertically disposed on the base, the camera bracket is disposed in front of the control board, the camera is fixedly disposed on the camera bracket, the camera and the window are correspondingly provided, and the control board is electrically connected to the camera and the wire harness, respectively.

(12) Optionally, the sunshading part is convex, and the sunshading part is provided with a plurality of heat dissipation holes.

(13) Optionally, the image acquisition apparatus further includes a decorative cover plate; the decorative cover plate is detachably disposed above the sunshading part.

(14) Optionally, the image acquisition apparatus further includes a front cover; the front cover is provided with a through-hole; the front cover is disposed in front of the upper box body, the upper box body is detachably connected with the front cover, and the through-hole and the window are correspondingly provided.

(15) Optionally, the image acquisition apparatus further includes a heat-conducting plate; the heat-conducting plate is disposed between the base and the control board.

(16) Optionally, the image acquisition apparatus further includes a heat dissipation part; the heat dissipation part includes a plurality of second ribs, and the surface of the base away from the camera assembly is vertically disposed with the plurality of second ribs at predetermined intervals, and the plurality of second ribs are configured to increase the surface area of the base. According to the second aspect of the present application, it is provided a self-moving device including an image acquisition apparatus as described above.

(17) For the image acquisition apparatus of the present application, the sunshading part is disposed at the side of the upper box body away from the base, and the sunshading part is disposed spaced from the upper box body, and the sunshading part is used to shade the sunlight to avoid the sunlight from directly irradiating the upper box body, thereby reducing light energy absorbed by the upper box body, and reducing temperature rise of the camera assembly inside the upper box body. On the other hand, the upper box body and the sunshading part are disposed spaced from each other, and an air flow layer is formed between the upper box body and the sunshading part. The air flow layer is blocked between the sunshading part and the upper box body, thereby preventing the transfer of heat from the sunshading part to the upper box body, and thus reducing temperature rise of the camera assembly inside the upper box body. Further, the air flowing inside the air flow layer takes away heat from the upper box body and the sunshading part, thereby reducing temperature rise of the camera assembly inside the upper box body. The image acquisition apparatus of the present application solves the problem of high temperature rise of the image acquisition apparatus and enhances the working efficiency of the image acquisition apparatus.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is an overall exploded schematic diagram of an image acquisition apparatus provided in accordance with the present application.

(2) FIG. 2 is an overall structure schematic diagram of an image acquisition apparatus provided in accordance with the present application.

(3) FIG. 3 is a schematic diagram of a combination of a base and a heat dissipation part in an image acquisition apparatus provided in accordance with the present application.

(4) FIG. 4 is a schematic diagram of an upper box body with a first rib structure in an image

acquisition apparatus provided in accordance with the present application.

(5) FIG. 5 is a schematic diagram of a combination of a base and a camera assembly in an image acquisition apparatus provided in accordance with the present application.

(6) FIG. 6 is a schematic diagram of a sunshading part in an image acquisition apparatus provided in accordance with the present application.

(7) FIG. 7 is a schematic diagram of a sunshading part with bolts in an image acquisition apparatus provided in accordance with the present application.

(8) FIG. 8 is a schematic diagram of a combination of an upper box body and a sunshading part in an image acquisition apparatus provided in accordance with the present application.

(9) FIG. 9 is a cross-sectional view of a combination of an upper box body and a sunshading part in an image acquisition apparatus provided in accordance with the present application.

(10) FIG. 10 is a schematic diagram of a combination of an image acquisition apparatus and a vehicle body in a self-moving device provided in accordance with the present application.

DETAILED DESCRIPTION OF EMBODIMENTS

(11) In order to make the objectives, technical solutions and advantages of the present application clearer, the following will be described in further detail on embodiments of the present application in conjunction with the accompanying drawings.

(12) In the description of the present application, it needs to be understood that the terms “center”, “up”, “down”, “front”, “back”, “left”, “right”, “vertical”, “horizontal”, “top”, “bottom”, “inside”, “outside”, and the like indicate the orientation or positional relationship based on that shown in the accompanying drawings. This is intended only to facilitate and simplify the description of the present application, and is not intended to indicate or imply that the apparatus or component referred to must have a particular orientation, be constructed and operated in a particular orientation, and therefore is not to be construed as a limitation of the present application.

(13) In the description of the present application, it should be illustrated that, unless otherwise expressly specified and limited, the terms “mount”, “connection”, and “connected” are to be understood in a broad sense, for example, the connection can be a fixed connection, a detachable connection, or an integrated connection; a mechanical connection or an electrical connection; a direct connection or an indirect connection through an intermediate medium; or an inner-connection within two components. To a person of ordinary skill in the art, the specific meaning of the above terms in the present application can be understood in specific cases.

(14) Referring to FIGS. 1 to 2, the present application provides an image acquisition apparatus 10 that can solve the problem of high temperature rise of the image acquisition apparatus 100 and enhance working efficiency of the image acquisition apparatus 100.

(15) The image acquisition apparatus 100 provided in the present application includes a base 10, an upper box body 20, a camera assembly 30, and a sunshading part 40, the upper box body 20 is detachably disposed on the base 10, a cavity is formed between the upper box body 20 and the base 10, and a front end face of the upper box body 20 is provided with a window 21; the camera assembly 30 is fixedly disposed on the base 10 and is placed in the cavity, and the camera assembly 30 and the window 21 are correspondingly provided; the sunshading part 40 is disposed at a side of the upper box body 20 away from the base 10, and the sunshading part 40 is disposed spaced from the upper box body 20.

(16) The base 10 and the upper box body 20 of the image acquisition apparatus 100 provided in the present application form the cavity for placing the camera assembly 30, and the base 10 is used to fix the camera assembly 30, and the sunshading part 40 is disposed at the side of the upper box body 20 away from the base, and the sunshading part 40 is disposed spaced from the upper box body 20. The sunshading part 40 is configured to shade the sunlight to avoid the sunlight from directly irradiating the upper box body 20, thereby reducing light energy absorbed by the upper box body 20, and thus reducing temperature rise of the camera assembly 30 inside the upper box body 20. On the other hand, the upper box body 20 is disposed spaced from the sunshading part 40, and

an air flow layer is formed between the upper box body **20** and the sunshading part **40**. The air flow layer is blocked between the sunshading part **40** and the upper box body **20**, thereby preventing the transfer of heat from the sunshading part **40** to the upper box body **20**, and reducing temperature rise of the camera assembly **30** inside the upper box body **20**. Further, the air flowing inside the air flow layer takes away heat from the upper box body **20** and the sunshading part **40**, thereby reducing temperature rise of the camera assembly **30** inside the upper box body **20**. The problem of high temperature rise of the image acquisition apparatus **100** is solved and the working efficiency of the image acquisition apparatus **100** is enhanced.

(17) The window **21** is of an external expansion structure to increase shooting angle of the camera assembly **30**.

(18) Optionally, the base **10** is detachably connected with the upper box body **20**, the camera assembly **30** is detachably connected with the base **10**, and the upper box body **20** is detachably connected with the sunshading part **40**.

(19) Further, the base **10** and the upper box body **20** can be connected with screws, the camera assembly **30** and the base **10** can be connected with screws, and the upper box body **20** and the sunshading part **40** can be connected with screws, which are not limited herein.

(20) Optionally, the material of the base **10** is metal.

(21) Further, the material of the base **10** is aluminum alloy, aluminum alloy has a higher thermal conductivity, which improves heat dissipation efficiency and facilitates reduction of temperature rise of the image acquisition apparatus **100**.

(22) Preferably, a surface of the sunshading part **40** near the upper box body **20** is provided with a plurality of bolts **41**, and a surface of the upper box body **20** near the sunshading part **40** is provided with a plurality of bolt holes **22**.

(23) Further, screw spikes pass through the bolts **41** from the upper surface of the sunshading part **40** and are connected to the bolt holes **22** respectively to fix the sunshading part **40** to the upper box body **20**. The bolts **41** are disposed protruding from the upper surface of the sunshading part **40**. When the sunshading part **40** is fixed to the upper box body **20**, lower surfaces of the bolts **41** are abutted against peripheries of the bolt holes **22**, respectively, so that the sunshading part **40** is disposed spaced from the upper box body **20**, and the air flow layer is formed between the sunshading part **40** and the upper box body **20**. The air flow layer is blocked between the sunshading part **40** and the upper box body **20**, thereby preventing the transfer of heat from the sunshading part **40** to the upper box body **20**, and reducing temperature rise of the camera assembly **30** inside the upper box body **20**. Further, the air flowing inside the air flow layer takes away heat from the upper box body **20** and the sunshading part **40**, thereby reducing temperature rise of the camera assembly **30** inside the upper box body **20**. The image acquisition apparatus **100** of the present application solves the problem of high temperature rise of the image acquisition apparatus **100**.

(24) Referring to FIGS. **3** to **5**, preferably, a lower surface of the base **10** is provided with a wiring hole **11**, and the wiring hole **11** is used for a wire harness to pass through, and a rubber plug **13** is disposed between the wire harness and the wiring hole **11**.

(25) The rubber plug **13** is disposed between the wire harness and the wiring hole **11**, so that there is no gap between the wire harness and the wiring hole **11** for waterproofing.

(26) Optionally, the lower surface of the base **10** is provided with a test hole, and the test hole is used for extracting air from the interior of the cavity.

(27) The test hole is provided to test waterproof performance of the image acquisition apparatus **100** provided in the present application by using an air pump to extract air from the interior of the cavity through the test hole, and after the test is completed, a patch is attached to the test hole by double-sided adhesive for sealing, in addition, a waterproof tape can also be attached to the test hole for sealing, which is not limited herein.

(28) Referring to FIGS. **4** to **5**, preferably, an upper surface of the base **10** is provided with a circle

of closed groove **14**. An inner surface of the upper box body **20** is vertically provided with a circle of closed first rib **23**. The first rib **23** is embedded in the groove **14** for interference fit. The first rib **23**, the groove **14**, the base **10**, and the upper box body **20** enclose to form a first cavity, and the camera assembly **30** is placed in the first cavity.

(29) The first rib **23** is embedded in the groove **14** for interference fit, which is in order to prevent water from entering the first cavity, so as to avoid water from entering the camera assembly **30** to make the camera assembly **30** not work properly. The shapes of the first rib **23** and the groove **14** can be circular or elliptic, and the shapes of the first rib **23** and the groove **14** in the present application are matched, which is not limited herein.

(30) Referring to FIGS. **4** to **5**, preferably, a waterproof ring **15** is provided in the groove **14**, and an upper surface of the first rib **23** is pressed against the waterproof ring **15**.

(31) The waterproof ring **15** is provided to play a waterproof role to prevent water from entering the interior of the first cavity.

(32) Referring to FIG. **8**, preferably, the upper box body **20** further includes a transparent part **24**, and the transparent part **24** is embedded on the window **21**.

(33) The transparent part **24** is provided to protect the camera assembly **30** from scratching. Optionally, an outer surface of the transparent part **24** is provided with a high-definition film or a blue light film, the high-definition film or blue light film can make shoot effect of the camera assembly **30** better as well as protect the outer surface of the transparent part **24** from scratching. Referring to FIGS. **1** to **5**, preferably, the camera assembly **30** includes a control board **31**, a camera bracket **32** and a camera **33**. The control board **31** is fixedly disposed on the base **10**, the camera bracket **32** is vertically disposed on the base **10**, the camera bracket **32** is disposed in front of the control board **31**, the camera **33** is fixedly disposed on the camera bracket **32**, the camera **33** and the window **21** are correspondingly provided, and the control board **31** is electrically connected to the camera **33** and the wire harness, respectively.

(34) The control board **31** and the camera bracket **32** can be disposed on the base **10** with screws. Referring to FIG. **5**, preferably, the control board **31** is disposed parallel to a bottom surface of the base **10**, the camera bracket **32** is disposed perpendicular to the bottom surface of the base **10**, the camera **33** is disposed on the camera bracket **32**, and the camera **33** and the control board **31** are designed non-parallel to reduce space sharing degree and reduce the influence of temperature rise of the control board **31** on the camera **33**.

(35) To further illustrate this embodiment, for example, when a self-moving device **200** works outdoors, it possible occur the situation of visual blind spots, and an image of a visual blind spot is taken by the camera **33** and transmitted to the control board **31**, and then is transmitted by the control board **31** through the wire harness to a display of the self-moving device **200** for a driver to perform real-time observation as well as real-time specific operation. For example, if the image capture device **100** is used in a harvester, the driver accurately harvests corn stalks in visual blind spots based on the uploaded images. The scenario is not limited herein.

(36) Optionally, the camera assembly **30** further includes a cable and a wireless transmission device; the control board **31** and the camera **33** are electrically connected by the cable or a wireless mode.

(37) Further, the cable is a flexible printed circuit cable (FPC cable), and the control board **31** and the camera **33** are electrically connected via the FPC cable or the wireless transmission device.

(38) Referring to FIGS. **6** to **9**, preferably, the sunshading part **40** is convex, and the sunshading part **40** is provided with a plurality of heat dissipation holes **42**.

(39) The sunshading part **40** is disposed in a convex shape and an outer end surface of the sunshading part **40** is provided with the plurality of heat dissipation holes **42**, thereby promoting air flowing and avoiding the accumulation of high temperature air between the sunshading part **40** and the upper box body **20**, this benefits to the internal heat dissipation of the image acquisition apparatus **100**.

(40) Referring to FIGS. 1 to 2, preferably, the image capture device **100** further includes a decorative cover plate **50**; the decorative cover plate **50** is detachably disposed above the sunshading part **40**.

(41) The sunshading part **40** can be covered by the decorative cover plate **50** in a snap-fit manner, and the decorative cover plate **50** is provided to cover the bolts in the sunshading part **40** to make the appearance of the image acquisition apparatus **100** more beautiful.

(42) Referring to FIGS. 1 to 2, preferably, the image capture device **100** further includes a front cover **60**; the front cover **60** is provided with a through-hole **61**; the front cover **60** is disposed in front of the upper box body **20**, the upper box body **20** is detachably connected with the front cover **60**, and the through-hole **61** and the window **21** are correspondingly provided.

(43) The front cover **60** is provided to protect the transparent part **24**, and the size of the through-hole **61** is smaller than or equal to that of the transparent part **24** to avoid the transparent part **24** from scratching and also to make the appearance of the image acquisition apparatus **100** more beautiful. The front cover **60** can be disposed on the upper box body **20** by a buckle, which is not limited herein.

(44) Referring to FIG. 1, preferably, the image acquisition apparatus **100** further includes a heat-conducting plate **70**; the heat-conducting plate **70** is disposed between the base **10** and the control board **31**. The surfaces of the control board **31**, the heat-conducting plate **70**, and the base **10** fit to each other to facilitate the transfer of heat from the control board **31** to the external air by means of solid heat conduction and to enhance heat dissipation rate. The image acquisition apparatus **100** of the present application solves the problem of high temperature rise of the image acquisition apparatus **100** and enhances the working efficiency of the image acquisition apparatus **100**.

(45) The heat-conducting plate **70** is provided for heat dissipation of the control board **31**. The heat-conducting plate **70** is a silicone heat-conducting plate. The silicone heat-conducting plate has a suitable softness and can well fit to the surfaces of the control board **31** and the base **10**. In addition, the silicone heat-conducting plate is chemically stable and will not contaminate components of the control board **31**.

(46) Referring to FIGS. 1 to 3, preferably, the image capture device **100** further includes a heat dissipation part **80**.

(47) The heat dissipation part **80** includes a plurality of second ribs **81**, and the surface of the base **10** away from the camera assembly **30** is vertically provided with a plurality of second ribs **81** at predetermined intervals, and the plurality of second ribs **81** are used to increase the surface area of the base **10**.

(48) The heat dissipation part **80** is disposed on the base **10** to increase the surface area of the base **10**, and the plurality of second ribs **81** are disposed vertically at predetermined intervals on the lower surface of the base **10** to make for solving the problem of high temperature rise of the image capture device **100** and to enhance the work efficiency of the image capture device **100**.

(49) Herein, the number of the second ribs **81** is not limited as long as the base **10** can achieve the best heat dissipation effect.

(50) Preferably, the image acquisition apparatus **100** is detachably disposed on the self-moving device **200**, and the image acquisition apparatus **100** is electrically connected to a main board in the self-moving device **200**.

(51) Referring to FIG. 10, the present application provides a self-moving device. The self-moving device **200** provided in the present application can solve the problem of high temperature rise of the image acquisition apparatus **100** and enhance the working efficiency of the image acquisition apparatus **100**.

(52) The self-moving device **200** includes the image acquisition apparatus **100** described above. The image acquisition apparatus **100** can be used to solve the problem of high temperature rise of the image acquisition apparatus **100** and enhance the working efficiency of the image acquisition apparatus **100**.

(53) Referring to FIG. 10, optionally, the self-moving device **200** further includes a vehicle body **90**, and the image acquisition apparatus **100** is detachably disposed on an outer surface of the vehicle body **90**.

(54) The image acquisition apparatus **100** is used to provide visual information to the vehicle body **90**, and when there is a visual blind spot, the image acquisition apparatus **100** can be used to avoid and locate.

(55) Optionally, the self-moving device **200** further includes a vehicle body **90**, and a surface of the heat dissipation part **80** away from the base **10** fits to the outer surface of the vehicle body **90**, i.e., surfaces of the plurality of second ribs **81** away from the base **10** fit to the outer surface of the vehicle body **90**.

(56) This embodiment is further illustrated, when the car body **90** is a mower, the surface of the heat dissipation part **80** away from the base **10** fits to the outer surface of the mower, i.e., the surfaces of the plurality of second ribs **81** away from the base **10** fit to the outer surface of the mower, thereby improving the stability of the image capture device **100** mounted on the mower.

(57) The image acquisition apparatus **100** is used to provide visual information to the vehicle body **90**, and when there is a visual blind spot, the image acquisition apparatus **100** can be used to assist, an image is taken by the camera **33** and transmitted to the control board **31**, and then is transmitted by the control board **31** to the main board of vehicle body **90**, and then is shown by a display for a driver to view and perform real-time specific operations. For example, if the image acquisition apparatus **100** is used in a harvester, the driver accurately harvests corn stalks in visual blind spots based on the uploaded images.

(58) Optionally, an outer end of the image acquisition apparatus **100** is provided with a protective shell at a predetermined spacing.

(59) The protective shell may be located on the vehicle body **90** or on the image acquisition apparatus **100**, which is not limited herein. When the image capture device **100** is used in a mower or a harvester machine, there will be spatter materials during the mower or harvester works, and the protective shell plays a protective role on the image acquisition apparatus **100**, and also prevents water from entering the image capture device **100** under raining.

(60) Scenario 1:

(61) When the image capture device **100** is used in a mower, the image capture device **100** is detachably mounted on the mower, and the mounting position on the mower depends on the position of the driver's visual blind spot. The mower generally works during the daytime, so that the image capture device **100** cannot avoid direct sunlight and the generation of heat itself when working. The image capture device **100** is provided with the sunshading part **40**, the sunshading part **40** is disposed at the side of the upper box body **20** away from the base **10**, and the sunshading part **40** is disposed spaced from the upper box body **20**, the sunshading part **40** is used to shade the sunlight and avoid the sunlight from directly irradiating the upper box body **20**, thereby reducing light energy absorbed by the upper box body **20**, and thus reducing temperature rise of the camera assembly **30** inside the upper box body **20**. On the other hand, the upper box body **20** is disposed spaced from the sunshading part **40**, and an air flow layer is formed between the upper box body **20** and the sunshading part **40**. The air flow layer is blocked between the sunshading part **40** and the upper box body **20**, thereby preventing transfer of heat from the sunshading part **40** to the upper box body **20**, and reducing temperature rise of the camera assembly **30** inside the upper box body **20**. Further, the air flow layer within the air flow layer takes away heat from the upper box body **20** and the sunshading part **40**, thereby reducing temperature rise of the camera assembly **30** inside the upper box body **20**. The sunshading part **40** is provided to solve the problem of high temperature rise of the image acquisition apparatus **100** and enhance the working efficiency of the image acquisition apparatus **100**.

(62) To sum up, the base **10** and the upper box body **20** of the image acquisition apparatus **100** provided in the present application form a cavity for placing the camera assembly **30**. The

sunshading part **40** is disposed spaced from the upper box body **20**, and the sunshading part **40** is used to shade the sunlight and avoid the sunlight from directly irradiating the upper box body **20**, thereby reducing the light energy absorbed by the upper box body **20**, and thus reducing temperature rise of the camera assembly **30** inside the upper box body **20**. On the other hand, the upper box body **20** is disposed spaced from the sunshading part **40**, and an air flow layer is formed between the upper box body **20** and the sunshading part **40**. The air flow layer is blocked between the sunshading part **40** and the upper box body **20**, thereby preventing the transfer of heat from the sunshading part **40** to the upper box body **20**, and reducing temperature rise of the camera assembly **30** inside the upper box body **20**. Further, the air flow layer within the air flow layer takes away heat from the upper box body **20** and the sunshading part **40**, thereby reducing temperature rise of the camera assembly **30** inside the upper box body **20**. The present application solves the problem of high temperature rise of the image acquisition apparatus **100** and enhances the working efficiency of the image acquisition apparatus **100**.

(63) It should be illustrated that not all the steps and modules in respective processes and respective system structure diagrams above are necessary, and certain steps or modules can be ignored according to actual needs. The order of execution of respective steps is not fixed and can be adjusted as needed. The system structure described in the aforesaid examples may be physical structure or logical structure, i.e., some modules may be implemented by a same physical entity or some modules may be implemented by multiple physical entities, respectively, or may be implemented by certain components of multiple independent devices together.

(64) In each of the examples above, a hardware module may be implemented by mechanical means or electrical means. For example, a hardware module may include permanently dedicated circuitry or logic (e.g., a specialized processor, FPGA or ASIC) to perform corresponding operations. A hardware module may also include programmable logic or circuitry (e.g., a general-purpose processor or other programmable processors) that can be temporarily set by software to perform corresponding operations. The specific implementation means (mechanical means, or dedicated permanent circuitry, or temporary set circuitry) can be determined based on cost and time considerations.

(65) The present application is presented and illustrated in detail above by the accompanying drawings and preferred examples. However, the present application is not limited to these disclosed examples, based on the aforesaid multiple examples, it is known to persons of ordinary skill in the art that more examples of the present application can be obtained by combining the code review means in the various examples described above, and these examples are also within the scope of protection of the present application.

Claims

1. An image acquisition apparatus, comprising: a base; an upper box body, the upper box body is detachably disposed on the base, a cavity is formed between the upper box body and the base, and a front end face of the upper box body is provided with a window; a camera assembly fixedly disposed on the base and placed in the cavity, and the camera assembly and the window are correspondingly provided; a sunshading part disposed at a side of the upper box body away from the base, and the sunshading part is disposed spaced from the upper box body; the camera assembly comprises a control board, a camera bracket, and a camera; and the control board is fixedly disposed on the base, the camera bracket is vertically disposed on the base, the camera bracket is disposed in front of the control board, the control board is disposed parallel to a bottom surface of the base, the camera is disposed non-parallel to the control board.

2. The image acquisition apparatus according to claim 1, wherein a surface of the sunshading part near the upper box body is provided with a plurality of bolts, and a surface of the upper box body near the sunshading part is provided with a plurality of bolt holes.

3. The image acquisition apparatus according to claim 1, wherein a lower surface of the base is provided with a wiring hole for a wire harness to pass through, and a rubber plug is disposed between the wire harness and the wiring hole.
 4. The image acquisition apparatus according to claim 3, wherein an upper surface of the base is provided with a circle of closed groove, an inner surface of the upper box body is vertically provided with a circle of closed first rib, the first rib is embedded in the groove for interference fit, and the first rib, the groove, the base, and the upper box body enclose to form a first cavity, and the camera assembly is placed in the first cavity.
 5. The image acquisition apparatus according to claim 4, wherein a waterproof ring is disposed in the groove, and an upper surface of the first rib is pressed against the waterproof ring.
 6. The image acquisition apparatus according to claim 1, wherein the upper box body further comprises a transparent part, and the transparent part is embedded on the window.
 7. The image acquisition apparatus according to claim 3, wherein the camera is fixedly disposed on the camera bracket, the camera and the window are correspondingly provided, and the control board is electrically connected to the camera and the wire harness, respectively.
 8. The image acquisition apparatus according to claim 1, wherein the sunshading part is convex, and the sunshading part is provided with a plurality of heat dissipation holes.
 9. The image acquisition apparatus according to claim 1, wherein the image acquisition apparatus further comprises a decorative cover plate; and the decorative cover plate is detachably disposed above the sunshading part.
 10. The image acquisition apparatus according to claim 1, wherein the image acquisition apparatus further comprises a front cover; the front cover is provided with a through-hole; and the front cover is disposed in front of the upper box body, the upper box body is detachably connected with the front cover, and the through-hole and the window are correspondingly provided.
 11. The image acquisition apparatus according to claim 7, wherein the image acquisition apparatus further comprises a heat-conducting plate; and the heat-conducting plate is disposed between the base and the control board.
 12. The image acquisition apparatus according to claim 7, wherein the image acquisition apparatus further comprises a heat dissipation part; and the heat dissipation part comprises a plurality of second ribs, and the lower surface of the base away from the camera assembly is disposed with the plurality of second ribs at predetermined intervals, and the plurality of second ribs are configured to increase surface area of the base.
 13. A self-moving device comprising the image acquisition apparatus according to claim 1.
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